Persona Corpus C

166 if-then atoms · ESM-mined · Beck & Jackson (2022)

CN-high | CE-high | CC-high | CA-high | CO-high

CN-low | CE-low | CC-low | CA-low | CO-low

$\pi_{d,p}$ mixture weights *(deployer)*

$z_d \sim \text{Cat}(\pi_d)$

(query at inference time) *(from corpus)*

Situation s
"a friend cancels"

Selected cells $\{C_z\}$

Retriever
all-MiniLM-L6-v2 · cosine · TopK ($k = 4$)
situation-only encoding (§3)

Retrieved atoms $\sigma(R)$ → system prompt

- `WHEN a small inconvenience hits → I dwell, ruminate`
- `WHEN plans change unexpectedly → my mood drops`
- `WHEN alone in the evening → I replay it in my head`
- `WHEN unsure → I assume the worst`

↳ *no trait labels in the prompt — personality emerges from the atoms*

$\sigma(\cdot)$ *render*

LLM (T = 0.7)
MiniMax · GPT-4o · Gemini · Qwen · L8B

Response y
(scored on Mini-IPIP / PB)

Audit trail
fired atoms →
ESM provenance row

$$z_d \sim \text{Cat}(\pi_d), \quad R \sim P_\tau(R \mid q, C_{z_d}) \propto e^{\text{sim}(q,R)/\tau}, \quad y \sim P_\theta(y \mid q, \sigma(R))$$